FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

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Supplementary Information

Table of contents:

1 FlowPilot workflow, data contracts, and run provenance

FlowPilot separates chemical interpretation, engineering calculation, equipment realization, and publication of executable instructions into explicit computational stages. This separation is intended to prevent an unconstrained language-model response from being presented directly as a laboratory procedure. The complete workflow begins with a standardized chemist-facing intake and ends with either a validated, inventory-assigned design or a diagnostic blocked state. The workflow is governed by an explicit information-authority order: measured experimental evidence, hard safety and inventory constraints, confirmed batch-protocol facts, chemist hypotheses, model inference. Consequently, historical measurements and declared laboratory limits constrain candidate generation and revision, whereas a hypothesis is retained as an idea to test rather than promoted to an observed fact.

System architecture and decision authority

A FlowPilot run passes through standardized intake, batch parsing, upstream chemistry analysis, plan-aware retrieval, deterministic calculation and design-space exploration, downstream proposal generation, multi-agent council review, inventory reconciliation, final validation, and artifact publication (**Figure S1A**). The upstream chemistry layer identifies the transformation, mechanistic features, stage structure, stream incompatibilities, phase behavior, and candidate process bottlenecks. The retrieval layer supplies literature analogies and engineering rules relevant to that plan. These model-derived interpretations inform design, but they cannot supersede confirmed protocol facts, measurements, or hard operating limits. Deterministic components calculate and reconcile dependent quantities such as flow rate, reactor volume, residence-time basis, gas delivery, pressure settings, and inventory assignments. The council can criticize a candidate and propose bounded revisions, but a council response is not an executable output. Revised fields are recalculated and checked before publication. The final disposition therefore depends on deterministic closure rather than model confidence. Three user-facing outcomes are distinguished. An executable screening design contains canonical parameters and a fully assigned process topology. An inventory-confirmation state identifies the missing capability required before numerical execution. A blocked state records the failed requirements and diagnostic topology but withholds executable run instructions.

Typed contracts and numerical closure

Major pipeline boundaries are represented by validated data objects rather than prose alone (**Figure S1B**). The principal chain is DesignInputPackageBatchRecord > ChemistryPlan > FlowProposal > DesignCalculations > inventory-assigned ProcessTopology > FinalDesignContract. Natural-language rationales may accompany these objects, but downstream code consumes typed fields and performs boundary validation.

The FinalDesignContract is rebuilt after engineering realization and inventory reconciliation. For an executable result, it contains canonical parameters, stage definitions, streams, instrument manifest, process graph, chemistry identity, component placement, safety controls, operating procedure, and validation experiments. If arithmetic, residence-time basis, chemistry identity, phase semantics, equipment allocation, topology, safety, procedure, or diagram consistency does not close, executable parameters remain absent from the published contract and preliminary values are retained only under a diagnostic section.

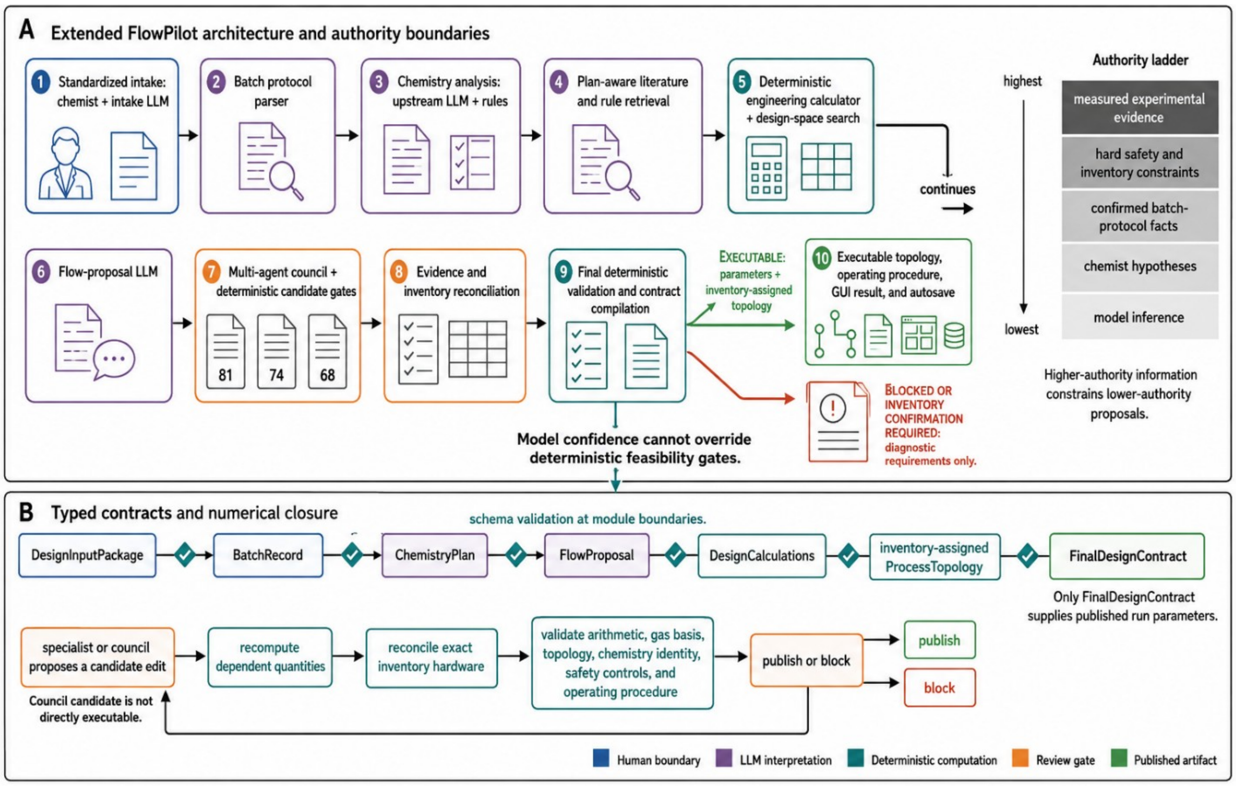

Figure S1. **FlowPilot architecture, authority boundaries, and typed data contracts.** (A) The end-to-end workflow separates chemist input, LLM interpretation, deterministic computation, review gates, and published artifacts. Information authority decreases from measured evidence and hard constraints to model inference. Final deterministic gates distinguish executable designs from inventory-confirmation and blocked diagnostic outputs. (B) Validated data objects are passed between modules. Candidate edits trigger recalculation, inventory reconciliation, and consistency checks before a FinalDesignContract can publish operating parameters.

Table S1. FlowPilot components, data contracts, and decision responsibilities.

Component	Computational class	Principal input	Principal output	Decision responsibility
Standardized intake	LLM extraction + deterministic state rules	Free-text protocol and chemist answers	DesignInputPackage	Structures user information; fixed IDs and readiness rules remain deterministic.
Batch protocol parser	Structured parser + schema validation	Frozen protocol facts	BatchRecord	Normalizes batch conditions; cannot outrank the original confirmed protocol.
Chemistry analysis	Upstream LLM + retrieved fundamentals	BatchRecord, intake context	ChemistryPlan	Interprets mechanism, stages, sensitivities, and stream logic; does not publish hardware settings.
Plan-aware retrieval	Deterministic filtering and ranking	ChemistryPlan and retrieval query	Ranked analogies and rules	Supplies supporting precedents; retrieved analogies do not override measured evidence or constraints.
Engineering calculator and design-space search	Deterministic	BatchRecord, ChemistryPlan, inventory limits	DesignCalculations and feasible seeds	Owns arithmetic closure, candidate feasibility calculations, and bounded search outputs.
Flow-proposal agent	Downstream LLM	Chemistry, calculations, retrieval, inventory context	FlowProposal	Proposes a complete candidate and rationale; proposal remains non-executable until validation.
Multi-agent council	Specialist LLMs + deterministic gates	Candidate proposals and evidence	Selected or revised candidate	Critiques chemistry, engineering, safety, and practicality; revisions are bounded and recalculated.
Inventory reconciliation and topology compiler	Deterministic	Selected candidate and LabInventory	Inventory allocation and ProcessTopology	Binds required operations to declared equipment and rejects unresolved assignments.
Final validation and artifact compiler	Deterministic	Reconciled proposal, calculations, topology, safety context	FinalDesignContract	Owns executable/blocked disposition and canonical operating parameters.
GUI renderer and autosave	Deterministic presentation	FinalDesignContract plus	Result views and stored	Publishes canonical outputs; diagnostic

Component	Computational class	Principal input	Principal output	Decision responsibility
	and serialization	diagnostic traces	artifacts	information is labelled and kept separate from run instructions.

Reproducible standardized intake

The intake layer converts an initial free-text protocol and follow-up answers into a frozen DesignInputPackage. The LLM may extract batch fields and conversationally present missing questions, but it selects from a fixed bank of stable identifiers. It cannot invent a new question identifier or mark an incomplete package as ready. The current question bank is listed in **Table S2**. The batch protocol, design objective, and chemistry identity must be answered. Historical data, inventory, operating limits, and hypotheses must either be answered or explicitly marked unavailable. Output preference is optional. Readiness and the ordered list of pending identifiers are calculated from package state. The frozen package records the raw protocol, extracted batch fields, objective, historical evidence, inventory constraints, operating limits, hypotheses, output preferences, chemistry confirmation, question log, answers, readiness state, and authority order. This design makes the intake boundary reproducible without claiming that every generative phrase is identical across model calls. Reproducibility applies to the allowed question identifiers, resolution rules, serialized chemist responses, readiness decision, and frozen context delivered to downstream modules. Where exact computational replay is required, the frozen package must be combined with model, decoding, software-revision, and environment records.

Table S2. Fixed standardized-intake question bank and readiness rules.

Information category	Readiness role	Permitted resolution	Downstream use
Batch protocol	Required	Answered only	Establishes confirmed protocol facts and batch conditions.
Design objective	Required	Answered only	Defines how candidates are ranked and what the first design should achieve.
Chemistry identity	Required	Answered only	Confirms transformation family, substrate/product identity, and bond changes.
Historical experiments	Required resolution	Answered or unavailable	Provides measured evidence for closed-loop calibration and trend constraints.
Laboratory inventory	Required resolution	Answered or unavailable	Supplies hard equipment capabilities and discrete

Information category	Readiness role	Permitted resolution	Downstream use
			hardware options.
Operating limits	Required resolution	Answered or unavailable	Defines pressure, temperature, flow, material, and safety boundaries.
Chemist hypotheses	Required resolution	Answered or unavailable	Guides candidate screening but is explicitly treated below measured evidence and confirmed facts.
Output preference	Optional	Answered or omitted	Controls presentation and requested screening format without changing physical feasibility rules.

GUI execution and single-source result rendering

In the standardized-intake GUI, the user submits a protocol, resolves the fixed intake questions, and imports or selects a laboratory-inventory profile before initiating design. The run action is enabled only when the package is ready. The finalized package is passed to the same translation pipeline used by programmatic execution, and the resulting package is attached to the result. After final validation, executable numerical views are constructed from the post-realization FinalDesignContract. These include the summary, engineering design, stream assignments, equipment allocation, process graph, safety controls, and canonical operating procedure. Chemistry-plan, retrieval, council, and experiment-loop records remain available as supporting context or audit traces. They are not independently promoted as authoritative run parameters. For a blocked run, the GUI displays the failure reasons and a requirements or diagnostic topology while withholding an executable procedure.

Run-level provenance and reconstruction

Completed GUI runs are automatically written to a uniquely timestamped directory. The artifact set preserves the standardized input, complete result, canonical final contract, concise summary, compiled topology, equipment allocation, instrument manifest, and available rendered diagrams (**Table S3**). Diagnostic topology and rendering artifacts are stored separately when the run is blocked or requires inventory confirmation. The render manifest can associate diagram files with topology-hash metadata, allowing a stored diagram to be checked against the corresponding graph where that metadata is available. The autosave package supports inspection of what was entered, calculated, validated, displayed, and assigned to hardware. It should not, by itself, be described as a complete computational-replay bundle because separate prompt logs, exact model snapshots, decoding parameters, dependency versions, and software revision identifiers are not guaranteed by the GUI autosave function.

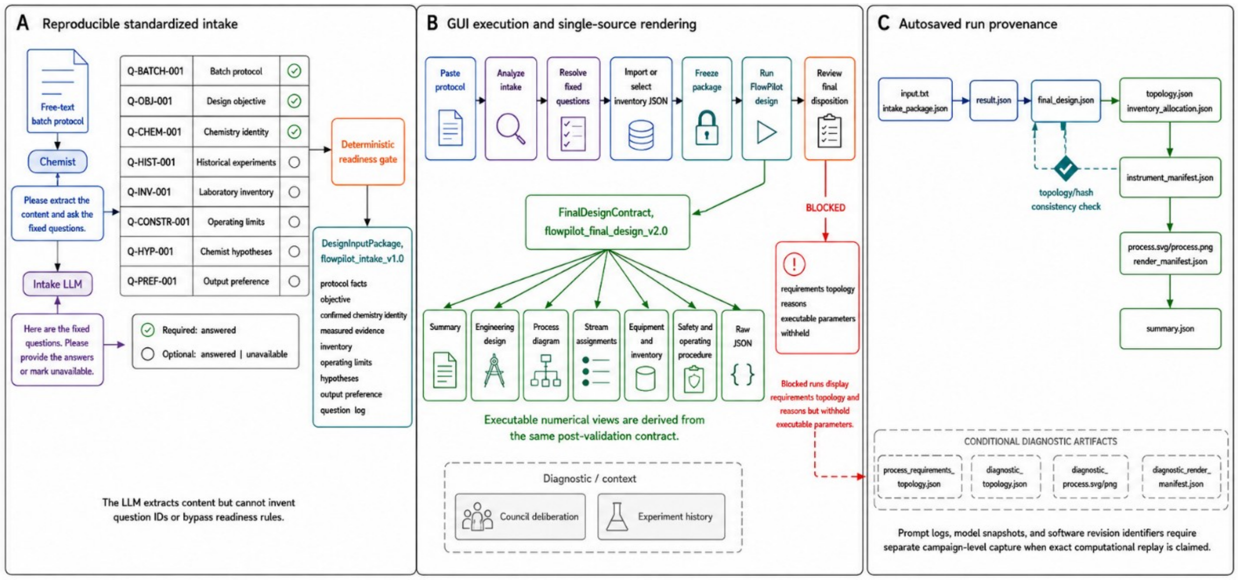

Figure S2. Reproducible standardized intake, GUI execution, and run provenance. (A) A conversational intake LLM extracts content while fixed question identifiers and deterministic readiness rules control completion of the DesignInputPackage. (B) The frozen package and selected inventory enter the design pipeline. Executable numerical views are derived from the same post-validation FinalDesignContract, whereas blocked runs expose requirements and reasons without run parameters. (C) Autosaved artifacts link user input and intake context to the complete result, canonical final design, inventory-assigned topology, instrument manifest, rendered process scheme, and compact summary. Diagnostic files are stored separately from executable artifacts.

Table S3. GUI autosave artifacts and their provenance function.

Artifact	Creation condition	Provenance function
input.txt	When raw user input is supplied to autosave	Original GUI text submitted for the run.
intake_package.json	When an intake package is used	Frozen standardized context, answers, readiness state, and authority order.
result.json	Every completed autosave	Complete JSON-safe pipeline result, including available intermediate and diagnostic records.
final_design.json	When the final contract is available	Canonical executable or blocked FinalDesignContract used by numerical result views.
summary.json	Every completed autosave	Compact status and principal design fields for indexing and review.
topology.json	When a process topology is compiled	Stored unit-operation graph and stream connections.
inventory_allocation.json	When inventory	Equipment assignments, unresolved

Artifact	Creation condition	Provenance function
	allocation is attempted	requirements, and allocation status.
instrument_manifest.json	When a manifest is produced	Declared instruments assigned to the final design.
process.svg; process.png	For a rendered executable topology	Portable visual representation of the canonical process graph.
render_manifest.json	When executable rendering is attempted	Rendering status, artifact paths, and topology-hash metadata where available.
process_requirements_topology.json	Conditional: unresolved inventory preflight	Required process structure before complete equipment assignment.
diagnostic_topology.json; diagnostic_process.svg/png; diagnostic_render_manifest.json	Conditional: blocked or diagnostic run	Non-executable requirements view and rendering diagnostics; not operating instructions.

2 Literature corpus, rule base, and retrieval

FlowPilot uses two complementary knowledge resources. A structured literature corpus provides empirical batch-to-flow precedents and process metadata, whereas a handbook-derived rule base provides general engineering principles. Plan-aware retrieval selects a small set of literature analogies for the chemistry under consideration, and the upstream chemistry layer receives a prioritized subset of relevant handbook rules. These resources provide evidence and context; they do not replace deterministic calculations, laboratory inventory constraints, or experimental validation.

Frozen literature-corpus composition

The frozen manuscript export contains 464 classified flow-chemistry literature records. The classification spans thermal synthesis, photoredox and heterogeneous photocatalysis, cross-coupling, oxidation and reduction, electrochemistry, biocatalysis, polymer synthesis, crystallization, and other continuous-flow applications. For each record, available information was normalized into fields describing reaction class, reactor type, reactor material, bond or transformation label, operating conditions, number of inlet streams, and reported batch and flow outcomes. The primary categorical exports each contain 464 classified entries, but field completeness is not uniform. Explicit inlet-stream counts are available for 154 processes, batch yields for 135 records, and optimized-flow yields for 285 records (Table S4). Unknown, other, multiple, and long-tail classifications are retained rather than being excluded from denominators. This avoids inflating apparent coverage and makes the limitations of the extracted metadata visible.

The available yield distributions are descriptive. They should not be interpreted as a paired estimate of the causal benefit of flow because batch and flow values are missing for different subsets, publication and optimization selection can affect the distributions, and individual records differ in chemistry and objective. Figure S3 therefore reports field-specific denominators directly in each panel.

Figure S3. Composition of the frozen flow-chemistry literature corpus. Distributions are shown for reaction class, reactor type, reactor material, bond or transformation label, number of inlet streams, and available batch and optimized-flow yields. The four principal categorical exports contain 464 classified records. Inlet-stream and yield panels use their own available-value denominators. Unknown, other, multiple, and long-tail categories are retained. Yield distributions are descriptive and are not treated as a paired causal comparison.

Table S4. Frozen literature-corpus fields and available denominators.

Metadata field	Available n	Coverage	Leading categories or summary	Interpretation note
Reaction class	464	100% classified export	Thermal Synthesis 96 (20.7%); Photoredox Catalysis 85 (18.3%); Other 66 (14.2%); Heterogeneous Photocatalysis 35 (7.5%); Cross-Coupling 32 (6.9%)	Other and long-tail classes retained.
Reactor type	464	100% classified export	Capillary / Coil Reactor 122 (26.3%); Microreactor / Chip 121 (26.1%); Continuous-Flow Reactor 71 (15.3%); Photoreactor 47 (10.1%); Packed-Bed Reactor 27 (5.8%)	Long-tail hardware labels retained.
Reactor material	464	100% classified export	Fluoropolymer (PTFE / PFA / FEP) 272 (58.6%); Glass / Quartz 61 (13.1%); Stainless Steel 58 (12.5%); PDMS / Silicon 28 (6.0%)	Classification granularity varies across records.
Bond/ transformation label	464	100% classified export	C–C 123 (26.5%); Other / Multiple 84 (18.1%); C–N 70 (15.1%); C–O 62 (13.4%); C–S 22 (4.7%)	Other/multiple is an explicit category.
Explicit inlet-stream count	154	33.2% of 464	1 streams: 22 (14.3%); 2 streams: 91 (59.1%); 3 streams: 30 (19.5%); 4 streams: 5 (3.2%); 5 streams: 3 (1.9%)	Percentages use n = 154, not the full corpus.
Reported batch yield	135	29.1% of 464	Median 80%; IQR 56-93%	Available numeric values only.

Metadata field	Available n	Coverage	Leading categories or summary	Interpretation note
Reported optimized-flow yield	285	61.4% of 464	Median 89%; IQR 75-96%	Not a paired causal comparison with batch yield.

Engineering rule-base construction and use

The engineering knowledge store contains 2,537 structured rules extracted from six handbook, guide, and literature sources. Each stored rule includes a rule identifier, category, triggering condition, recommendation, reasoning, optional quantitative expression, exceptions, severity, confidence, source document, source page, and source context when available. The complete export contains 29 rule categories; the 14 largest are summarized in **Table S5**. Across the complete store, 1,772 rules are labelled guideline, 717 hard_rule, 47 tip, and 1 safety severity. Category and severity are distinct labels. For example, most entries in the Safety category are encoded as hard_rule rather than with the separate safety severity label. Figure S4 therefore displays both dimensions rather than treating the Safety category as a severity class. At run time, the knowledge store selects broad engineering categories from the reaction context and adds keyword matches from the mechanism, catalyst, solvent, phase regime, and operating conditions. Duplicate rule identifiers are removed, hard rules are prioritized, and a compact subset is formatted for the upstream chemistry prompt. The downstream pipeline receives the resulting ChemistryPlan, while executable numerical requirements remain the responsibility of deterministic calculators and validation gates. The rule-base labels should not be overinterpreted. A machine-detected quantitative expression indicates that a rule contains formula-like content; it does not demonstrate independent equation verification. Similarly, a hard_rule label controls prompt prioritization but is not automatically equivalent to an executable coded constraint. A rule becomes a deterministic gate only when the corresponding calculation or validation check is implemented in code.

Table S5. Largest engineering rule categories and severity composition.

Category	Total	Hard rule	Guideline	Tip	Safety
Reactor Design	403	98	295	10	0
General	347	100	240	7	0
Heat Transfer	227	48	175	4	0
Catalyst	226	33	190	3	0
Residence Time	204	50	152	2	0
Photochemistry	175	21	152	2	0
Materials	165	62	97	6	0
Scale-Up	157	31	121	5	0
Pressure	134	57	76	0	1
Mixing	133	48	84	1	0
Safety	122	93	29	0	0
Solvent	72	26	44	2	0
Mass Transfer	59	18	39	2	0
Temperature	38	17	21	0	0

Figure S4 | Engineering rule-base structure (2,537 rules)

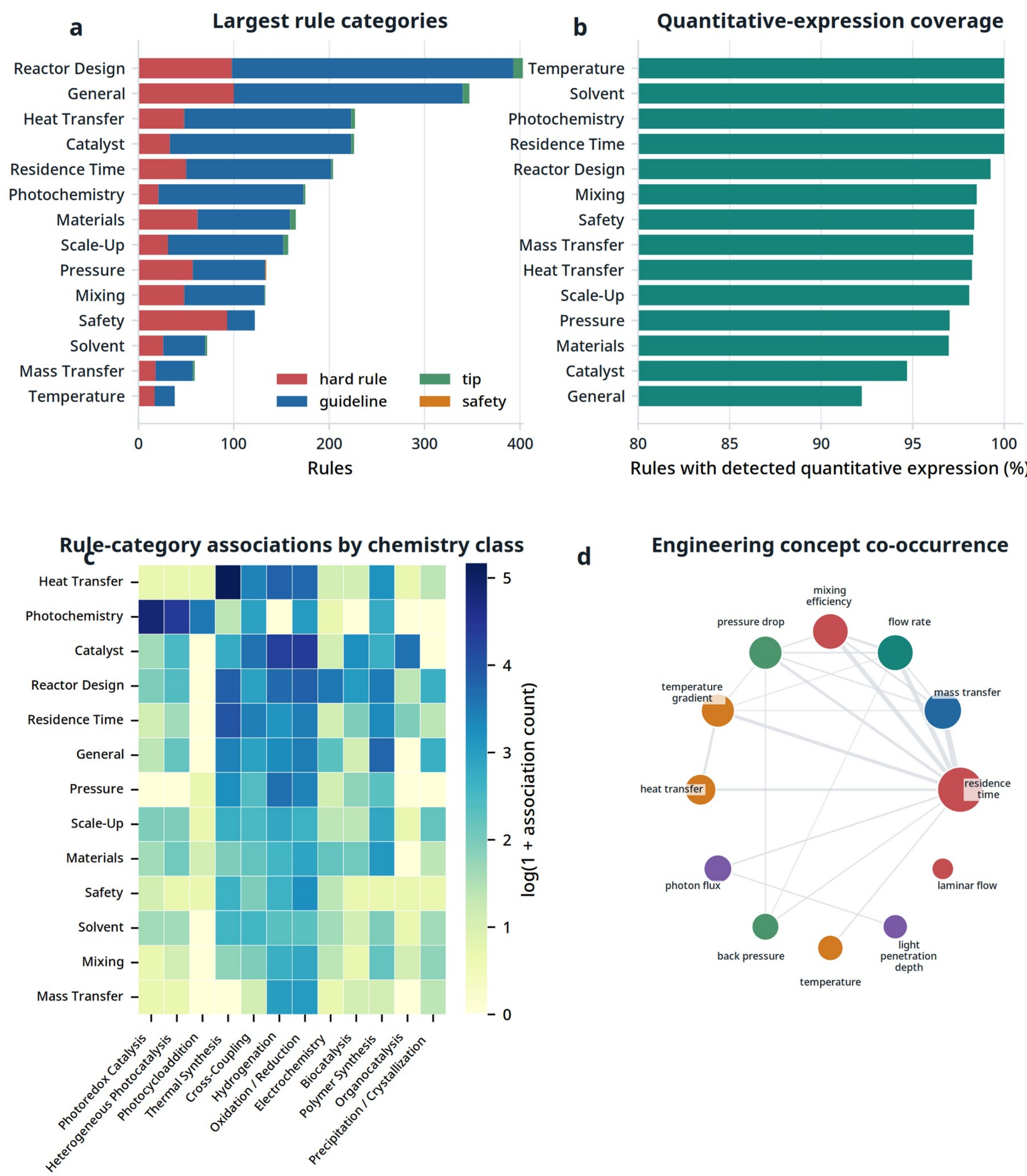

Figure S4. Engineering rule-base structure. (A) Counts by category and severity for the 14 largest categories. (B) Fraction of rules containing a machine-detected quantitative expression; this indicates expression coverage, not independent equation verification. (C) Non-exclusive rule-category associations across chemistry classes. (D) Co-occurrence network of frequent engineering concepts.

Plan-aware retrieval and tier transitions

The retrieval query is constructed from the ChemistryPlan rather than from protocol keywords alone. Reaction class, mechanism, catalyst or photocatalyst, solvent, phase regime, wavelength, temperature, and concentration are used when available. Retrieval begins with mechanism and phase filters over paired batch-flow records. If fewer than three candidates are found, the metadata filters are relaxed; if no paired records remain, the search expands to the complete corpus. Candidate records are reranked using 0.60 semantic similarity and 0.40 field similarity. The field score combines photocatalyst, solvent, wavelength, temperature, and concentration terms with weights of 0.30, 0.20, 0.20, 0.15, and 0.15, respectively (Figure S5).

Table S6. Frozen plan-aware retrieval tiers and reranking weights.

Component	Setting	Scope or transition
Tier 1	Paired records; mechanism + phase filters	Relax when fewer than three hits remain
Tier 2	Paired records; filters relaxed	Expand when no hits remain
Tier 3	All records; no metadata filter	Return ranked available records
Semantic similarity	0.60	Contribution to final reranking score
Field similarity	0.40	Contribution to final reranking score
Photocatalyst	0.30	Contribution within field similarity
Solvent / wavelength	0.20 / 0.20	Contributions within field similarity
Temperature / concentration	0.15 / 0.15	Contributions within field similarity

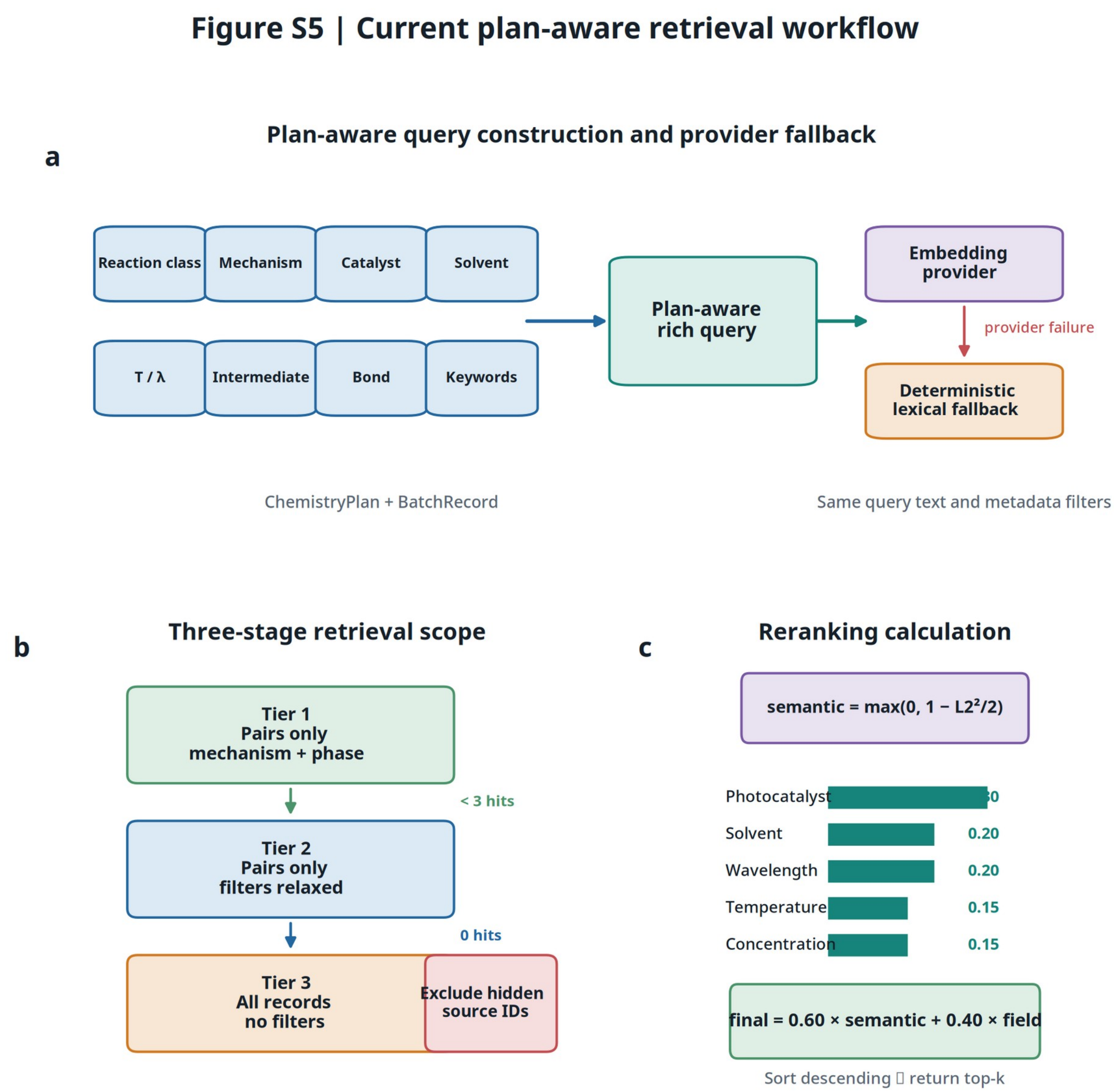

Figure S5. Plan-aware retrieval workflow. (A) ChemistryPlan fields enrich the query before semantic or deterministic lexical retrieval. (B) Tiered retrieval starts with mechanism and phase filters, relaxes filters when coverage is insufficient, and finally searches all records. (C) Semantic and field similarities are combined during reranking. Held-out source identifiers are excluded before final top-k selection.

Retrieval benchmark and leakage controls

The frozen retrieval analysis contains 1,600 query-result pairs. Reranking changed 25.1% of pairwise ranks. Top-five photocatalyst-family matching increased from 40.0% to 95.0% for iridium systems, 42.7% to 90.7% for organic dyes, 77.3% to 94.7% for ruthenium systems, 87.1% to 98.6% for TiO_2, and 45.0% to 60.0% for ZnO (Figure S6). These measurements quantify metadata alignment and retrieval behavior; they do not measure reaction yield, process optimality, or experimental success. For benchmark cases, the hidden source record identifier is removed after candidate construction and before top-k selection. This leave-one-source-out control prevents direct retrieval of the designated reference record but cannot establish absence from model pretraining.

Table S7. Frozen retrieval benchmark summary.

Metric or family	**Semantic retrieval**	**Plan-aware reranking**	**Evaluation n**
Query-result pairs reranked	-	25.1% changed rank	1,600 pairs
Iridium family match	40.0%	95.0%	8 queries
Organic-dye family match	42.7%	90.7%	15 queries
Ruthenium family match	77.3%	94.7%	15 queries
TiO_2 family match	87.1%	98.6%	14 queries
ZnO family match	45.0%	60.0%	4 queries

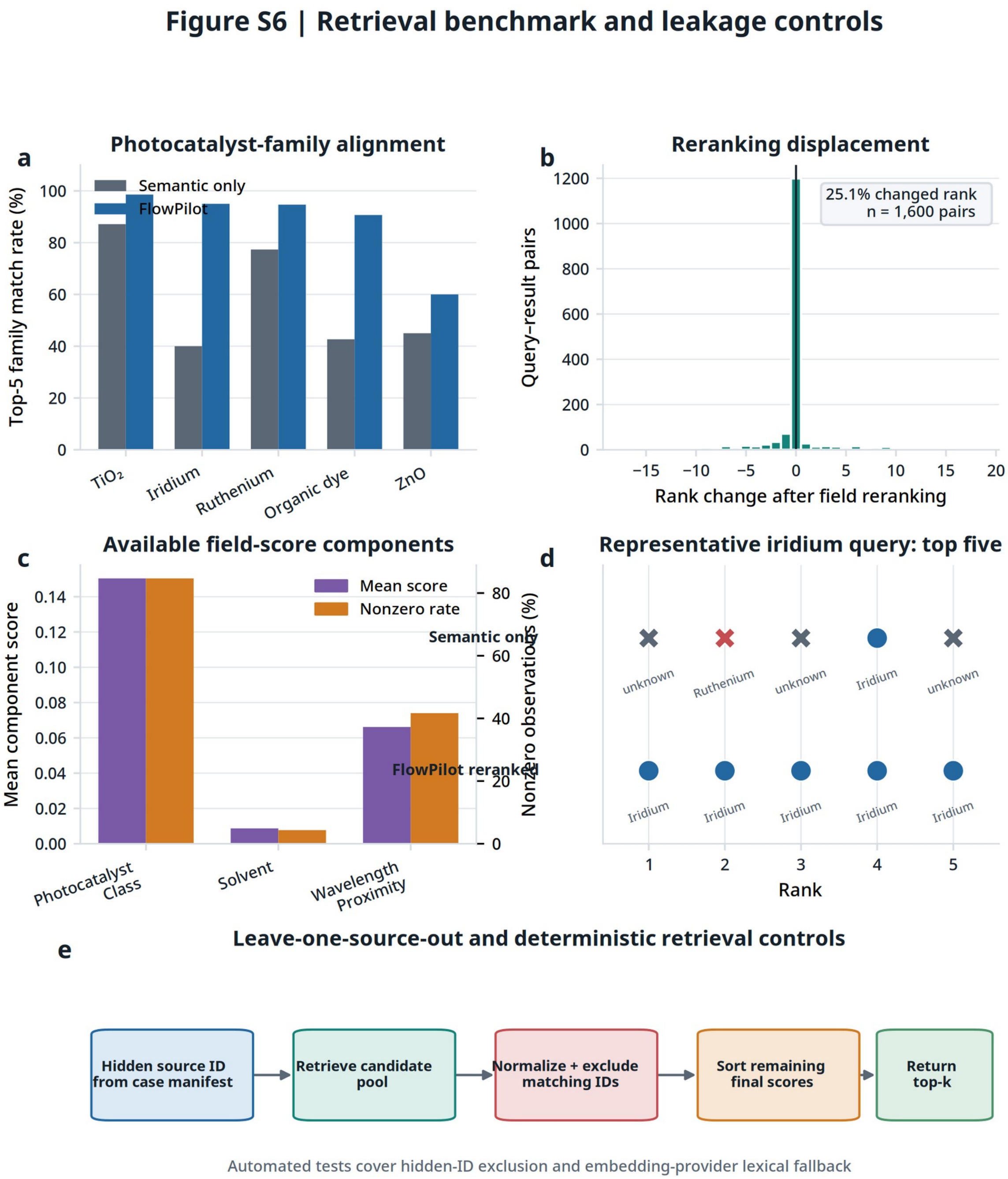

Figure S6. Retrieval benchmark and leakage controls. (A) Top-five photocatalyst-family match rates for semantic retrieval and FlowPilot reranking. (B) Rank changes across 1,600 frozen query-result pairs. (C) Field-score component availability. (D) Example iridium query before and after reranking. (E) Leave-one-source-out control and deterministic retrieval tests.

3 Ablation test and architecture comparison

Purpose and scope

The fixed-criteria multi-LLM evaluation was designed to compare the quality of final batch-to-flow designs produced by a direct one-shot large language model and by the complete FlowPilot architecture under matched inputs. The comparison was architecture-neutral at the evaluation stage: both methods received the same frozen chemistry, objective, laboratory inventory, and operating constraints, and both were required to return the same normalized final-design contract. The assessed object was the delivered design, not the internal reasoning trace or the number of intermediate agent calls. Two related experiments are reported in this section. The architecture comparison tested one-shot generation against full FlowPilot using five retained generator models, three chemistry cases, and three independent generation repeats. The internal module-ablation screen fixed the generator to Qwen3.8-27B and selectively removed or altered FlowPilot components to examine their contribution to outcome quality and executability.

The benchmark measures outcome quality and repeatability for the frozen cases and inventory. It does not establish universal superiority across all reaction classes and does not replace prospective wet-lab validation.

Paired architecture-comparison design

The retained manuscript comparison included the following generator models:

- Qwen3.6-27B
- Qwen3.8-27B
- GPT-4o
- Claude Sonnet 4.6
- Claude Opus 4.6

For every model, each of the three frozen chemistry cases was generated using both architectures in three independent calls. This produced:

5 models x 2 architectures x 3 cases x 3 repeats = 90 candidate outcomes

The paired unit was the same generator model, chemistry case, and repeat under the two architectures. The temperature was 0.2, calls were independently recorded, and the stopping rule was fixed before evaluation. The held-out source record for each case was excluded from FlowPilot retrieval. This exclusion prevents direct RAG leakage but does not prove that the underlying publication was absent from a model's pretraining data.

Compared architectures

One-shot. A single model call received the complete frozen design input and a fixed output contract. No FlowPilot specialist agents, council deliberation, deterministic candidate selection, or post-generation correction operated inside this architecture.

Full FlowPilot. The same frozen design input entered the complete FlowPilot workflow, including chemistry interpretation, engineering calculations, specialist-agent assessment, council selection and revision, safety analysis, deterministic closure checks, and final schema enforcement. The internal workflow could use several LLM calls, but only the final normalized design was scored.

Common one-shot wrapper

The one-shot system instruction was:

> *You are a general chemistry assistant. Translate a batch chemistry protocol into a practical continuous-flow experiment.*

For each chemistry, the corresponding frozen design input reported in Sections 1.5-1.7 was inserted after the header FROZEN DESIGN INPUT:. The following common response contract was then appended:

Exact one-shot response contract

> *Propose one complete flow design with numeric operating conditions, stream assignments, safety controls, and a brief rationale. Return one JSON object and no markdown with exactly these top-level fields: recommended_disposition, disposition_rationale, and proposal. The proposal value must be a JSON object, not a JSON-encoded string.*
>
> *The proposal object uses these fields:*
>
> *residence_time_min, flow_rate_mL_min, temperature_C, concentration_M, BPR_bar, reactor_type, tubing_material, tubing_ID_mm, reactor_volume_mL, residence_time_basis, residence_time_inlet_min, residence_time_in_channel_min, light_setup, wavelength_nm, streams, mixer_type, mixing_order_reasoning, pre_reactor_steps, post_reactor_steps, chemistry_notes, stage_parameters, reasoning_per_field, literature_analogies, engine_validated, safety_flags, confidence.*
>
> *Each stream is an object with stream_label, pump_role, contents, solvent, phase, flow_rate_mL_min, concentration_M, molar_equiv, gas_flow_sccm, and gas_flow_actual_mL_min where applicable. Distinguish inlet/STP gas flow from pressure-corrected in-channel gas flow. Ensure reactor_volume_mL is consistent with flow and residence time. State assumptions rather than inventing missing facts.*

recommended_disposition must be exactly EXECUTE, SCREEN, or BLOCK. Use BLOCK when the hard constraints make a safe design physically or operationally impossible. Use SCREEN for a feasible first experiment that still requires experimental validation. Use EXECUTE only for a directly executable design supported by sufficient evidence.

FlowPilot did not use one monolithic system prompt. It accepted the same frozen design input at its architecture boundary and distributed the information across its intake, chemistry, engineering, council, and validation stages. All internal prompts and stage events were retained in the archived run directories.

Identity-masked evaluation

Each final outcome was converted to a standardized candidate packet. Generator identity and architecture labels were removed before judging. Complete architectural blinding is not claimed because formatting or output style could reveal provenance. Every judge received the same protocol, objective, inventory, normalized final outcome, held-out reference facts, deterministic verification sheet, and fixed rubric.

Three independent judge families evaluated every candidate:

- Qwen
- OpenAI
- Claude

The deterministic sheet exposed machine-checkable evidence for schema validity, numerical closure, inventory compliance, reactor and residence-time consistency, gas bookkeeping where applicable, topology coverage, and safety requirements. Deterministic evidence informed the judges but did not itself add an architecture-specific score bonus.

Benchmark scoring method

The same 14 criteria were applied to every candidate. Each applicable criterion received an integer rating from 0 to 4. All criteria and all judges had equal weight; no architecture bonus or hand-tuned criterion weighting was used.

Table S8. Fixed 0-4 scoring anchors used by every independent judge.

Score	Anchor
0	Fundamentally wrong, unsafe, chemically invalid, or physically impossible.
1	Major deficiencies, multiple material defects, or poor reproducibility; substantial redesign is required.
2	Partially correct with one material but recoverable defect requiring revision before execution.
3	Correct overall with a minor defect that is unlikely to change execution or interpretation.
4	Correct, complete, executable, and supported; no material defect.

Table S9. Universal outcome criteria and applicability rules.

ID	Domain	Criterion	Applicability
UO-01	Chemistry	Transformation fidelity	Always
UO-02	Chemistry	Required materials	Always
UO-03	Chemistry	Stoichiometry and feed chemistry	Always
UO-04	Chemistry	Condition and stage mapping	Always
UO-05	Process	Executable topology	Always
UO-06	Engineering	Liquid material balance	Always
UO-07	Engineering	Residence-time and geometry closure	Always
UO-08	Engineering	Gas bookkeeping	Gas Reagent Or Gas Liquid Process
UO-09	Engineering	Multistage closure	More Than One Reactive Stage
UO-10	Inventory	Inventory feasibility	Always
UO-11	Engineering	Physical and transport plausibility	Always
UO-12	Safety	Hazards and controls	Always
UO-13	Operations	Operating procedure and work-up	Always
UO-14	Evidence	Evidence and uncertainty calibration	Always

NOT_APPLICABLE was permitted only when the criterion's applicability rule was false. Missing required information was rated 0 or 1 rather than excluded. In the frozen rubric, gas bookkeeping (UO-08) and multistage closure (UO-09) were the only criteria that could routinely become non-applicable. Critical-error flags were recorded separately and did not impose an arbitrary cap on the numerical score.

For candidate *i*, criterion *k*, and judge *j*, the criterion consensus was:

$$C_{i,k}=\frac{1}{J}\sum_{j=1}^{J} s_{i,k,j}$$

where s(i,k,j) is the judge's integer rating from 0 to 4. If A(i) is the set of applicable criteria, the candidate score was:

$$S_i=\frac{1}{|A_i|}\sum_{k\in A_i}\frac{C_{i,k}}{4}$$

The resulting candidate score ranged from 0 to 1. For each model, architecture, and repeat, the three chemistry candidate scores were first averaged to obtain a repeat-level campaign score:

$$R_{m,a,r}=\frac{1}{3}\sum_{q=1}^{3} S_{m,a,r,q}$$

The reported model-architecture value was the mean of the three repeat-level scores, and the error bar was their sample standard deviation (n = 3, ddof = 1). It was not a confidence interval. The matched architecture effect was calculated before aggregation:

$$\Delta_{m,q,r}=S_{FlowPilot,m,q,r}-S_{one\text{-}shot,m,q,r}$$

A positive value therefore favored FlowPilot for that matched case-repeat pair. Judge disagreement was retained separately and was not used as a substitute for generation-repeat variability.

Case study 1: Hydrogenolysis

Benchmark title: Three-phase N-diphenylmethyl hydrogenolysis

RAW BATCH PROTOCOL:

N-Diphenylmethylazetidin-3-ol (DMAOL; MW 239.32 g/mol) was prepared as a 3.8 wt% solution in methanol. For molarity conversion, explicitly use and label an assumed solution density of 0.792 g/mL, giving approximately 0.126 M DMAOL. A 200 mL charge was contacted with hydrogen over 0.5 g of 7 wt% Pd(OH)2/Al2O3 catalyst in a stirred batch reactor at 60 °C and 21 bar for 5 h. Conversion to 3-azetidinol was 38.1%. Translate this into one conservative continuous-flow packed-bed screening design. Acetic acid is optional and must not be introduced unless explicitly justified as a screening variable. Report feed preparation, gas-liquid mixing, packed-bed operation, temperature, pressure basis, liquid flow, hydrogen inlet/STP flow, pressure-corrected in-channel hydrogen flow, hydrogen equivalents, nominal empty-bed space time, liquid-holdup/contact residence-time basis, separation, collection, startup, shutdown, nitrogen purge, and hydrogen/methanol/pressure safety controls.

OBJECTIVE:

Produce one executable, inventory-constrained first gas-liquid-solid flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.

- Hydrogen is the required reducing reagent and must be delivered through the certified H2 MFC.

- Use a single DMAOL/methanol liquid feed and the declared gas-liquid mixer before the Pd(OH)2/Al2O3 packed bed.

- Operate at 60 °C and the available 21 bar BPR setpoint; explicitly state whether gas compression uses 21 bar absolute or 21 bar gauge.

- Report hydrogen flow at inlet/STP and pressure-corrected in-channel conditions, with the conversion calculation.

- Report H2 equivalents from gas molar flow divided by DMAOL molar flow.

- Distinguish nominal empty-bed space time from liquid-holdup/contact residence time; do not silently use total gas-plus-liquid flow as liquid reaction time.

- Use the nitrogen line for startup leak/purge verification and shutdown displacement of residual hydrogen.

- Return SCREEN unless the inventory cannot support a safe design.

AVAILABLE INVENTORY (JSON):

{"BPR_available":[21.0],"collectors":[{"equipment_id":"collector_h2_250","max_pressure_bar":1.0,"name":"Grounded vented liquid collection vessel","type":"collection vessel","volume_mL":250.0}],"connectors":[{"equipment_id":"connector_h2_pbr","material":"stainless steel","max_pressure_bar":40.0,"name":"Hydrogen-rated packed-bed connector set","supported_ID_mm":[4.35],"type":"packed-bed union"}],"degassers":[],"filters":[],"gas_hardware":[{"equipment_id":"gas_h2_cylinder","gas":"H2","max_pressure_bar":40.0,"name":"Certified hydrogen cylinder and regulator","type":"gas_supply"},{"equipment_id":"mfc_h2_100","gas":"H2","max_flow_sccm":100.0,"max_pressure_bar":40.0,"min_flow_sccm":1.0,"name":"Hydrogen mass-flow controller","type":"MFC"},{"equipment_id":"gas_n2_cylinder","gas":"N2","max_pressure_bar":40.0,"name":"Nitrogen purge cylinder and regulator","type":"gas_supply"}],"light_sources":[],"mixers":[{"equipment_id":"mixer_h2_t","material":"stainless steel","max_inputs":2,"max_pressure_bar":40.0,"max_total_flow_mL_min":5.0,"min_total_flow_mL_min":0.1,"name":"Hydrogen-service T-mixer","supported_ID_mm":[4.35],"type":"T-mixer"}],"pressure_controllers":[{"equipment_id":"bpr_h2_21","max_pressure_bar":21.0,"min_pressure_bar":21.0,"name":"Hydrogen-service BPR","setpoints_bar":[21.0],"type":"BPR"}],"pumps":[{"compatible_materials":["methanol","DMAOL"],"equipment_id":"pump_h2_pilot","max_flow_rate_mL_min":2.0,"max_pressure_bar":40.0,"min_flow_rate_mL_min":0.1,"name":"Pressure-rated plunger pump","type":"plunger"}],"reactor_trains":[],"reactors":[{"ID_mm":4.35,"allowed_temperatures_C":[60.0],"configuration":"single 200 mm gas-liquid-solid packed tube","equipment_id":"reactor_h2_pbr_300","material":"stainless steel","max_pressure_bar":40.0,"max_temperature_C":100.0,"min_pressure_bar":0.0,"min_temperature_C":20.0,"name":"Pd(OH)2/Al2O3 micro-packed bed","system":"hydrogenation skid","type":"packed-bed","volume_mL":3.0}],"safety_accessories":[{"capabilities":["backflow_prevention"],"compatible_hazards":["hydrogen"],"equipment_id":"safety_h2_check_valve","max_pressure_bar":40.0,"name":"Hydrogen non-return check valve","type":"check valve"},{"capabilities":["vented_separator"],"compatible_hazards":["hydrogen"],"equipment_id":"safety_h2_vent","max_pressure_bar":25.0,"name":"Dedicated hydrogen separator safe vent","type":"safe vent"}],"schema_version":"flowpilot_lab_inventory_v3.0","separators":[{"equipment_id":"separator_h2_gl","max_flow_mL_min":10.0,"max_pressure_bar":25.0,"name":"Vented gas-liquid separator","supported_phases":["gas","liquid"],"type":"gas-liquid separator"}],"strict_assignment":true,"temperature_controllers":[{"allowed_temperatures_C":[60.0],"equipment_id":"bath_h2_60","max_temperature_C":90.0,"min_temperature_C":20.0,"name":"Packed-bed water bath","type":"water bath"}],"tubing":[{"ID_mm":4.35,"equipment_id":"tube_h2_pilot_ss","length_m":0.2,"material":"stainless steel","max_pressure_bar":40.0,"max_temperature_C":100.0,"name":"Hydrogen-service packed tube","transparent":false}]}

CHEMIST-CONFIRMED CHEMISTRY IDENTITY:

{"confirmed":true,"mechanism_type":"heterogeneous catalytic hydrogenolysis","reaction_class":"hydrogenolysis","reaction_name":"N-diphenylmethyl hydrogenolysis"}

Case study 2: Photochemical oxidation

Benchmark title: Photocatalytic aerobic oxidation of Fmoc-L-methionine

RAW BATCH PROTOCOL:

Fmoc-L-methionine (3.7 mmol, 1.0 equiv) and 4,7-diphenyl-2,1,3-benzothiadiazole (10 mol%) were dissolved in acetonitrile at 0.10 M in a round-bottom flask. The solution was exposed to air and irradiated with a 1.5 W LED centered near 420 nm at 21 C for 5 min. Fmoc-methionine sulfoxide was formed in 99% yield. Translate this into one conservative continuous-flow photochemical screening design. Report feed preparation, air delivery and oxygen-equivalent basis, gas-liquid mixing, pressure basis, photoreactor assignment, wavelength, temperature, liquid flow, inlet/STP and pressure-corrected in-channel gas flow, liquid contact residence time, reactor volume, separation, collection, startup, shutdown, and oxygen/acetonitrile/light safety controls.

OBJECTIVE:

Produce one executable, inventory-constrained first gas-liquid photochemical flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.

- Use one premixed Fmoc-L-methionine/photocatalyst/acetonitrile liquid feed and the declared air MFC.

- Use the declared gas-liquid mixer before the 420 nm photoreactor.

- Operate at the available 3 bar BPR setpoint and state whether pressure is absolute or gauge.

- Report air at inlet/STP and pressure-corrected in-channel conditions and calculate oxygen equivalents from 21 mol% oxygen.

- Use liquid flow and the 1.0 mL reactor volume to define liquid contact residence time; do not use total gas-plus-liquid volumetric flow as reaction time.

- No inline degasser is available or required because air is a reagent.

- All selected equipment must be named by its inventory equipment_id.

- Return SCREEN unless the inventory cannot support a safe design.

AVAILABLE INVENTORY (JSON):

{"BPR_available":[3.0],"collectors":
[{"equipment_id":"collector_photo_100","max_pressure_bar":1.0,"name":"Amber vented collection vessel","type":"collection vessel","volume_mL":100.0}],"connectors":
[{"equipment_id":"connector_photo_set","material":"PFA","max_pressure_bar":10.0,"name":"PFA photochemistry connector set","supported_ID_mm":[1.0],"type":"photoreactor union set"}],"degassers":[],"filters":[],"gas_hardware":
[{"equipment_id":"gas_air_clean","gas":"air","max_pressure_bar":10.0,"name":"Clean dry air supply and regulator","type":"gas_supply"},
{"equipment_id":"mfc_air_10","gas":"air","max_flow_sccm":10.0,"max_pressure_bar":10.0,"min_flow_sccm":0.1,"name":"Air mass-flow controller","type":"MFC"}],"light_sources":
[{"allowed_temperatures_C":
[21.0,25.0],"compatible_reactor":"coil","equipment_id":"light_photo_420","max_temperature_C":40.0,"min_temperature_C":15.0,"name":"420 nm photochemistry LED","power_W":1.5,"wavelength_nm":420.0}],"mixers":
[{"equipment_id":"mixer_photo_t","material":"PFA","max_inputs":2,"max_pressure_bar":10.0,"max_total_flow_mL_min":2.0,"min_total_flow_mL_min":0.01,"name":"PFA gas-liquid T-mixer","supported_ID_mm":[1.0],"type":"T-mixer"}],"pressure_controllers":
[{"equipment_id":"bpr_photo_3","max_pressure_bar":3.0,"min_pressure_bar":3.0,"name":"Photochemistry BPR","setpoints_bar":[3.0],"type":"BPR"}],"pumps":[{"compatible_materials":
["acetonitrile","Fmoc-L-methionine","4,7-diphenyl-2,1,3-benzothiadiazole"],"equipment_id":"pump_photo_liquid","max_flow_rate_mL_min":1.0,"max_pressure_bar":10.0,"min_flow_rate_mL_min":0.01,"name":"Photochemistry HPLC pump","type":"HPLC"}],"reactor_trains":[],"reactors":
[{"ID_mm":1.0,"allowed_temperatures_C":[21.0,25.0],"configuration":"irradiated PFA coil","equipment_id":"reactor_photo_100","light_source":"light_photo_420","material":"PFA","max_pressure_bar":10.0,"max_temperature_C":40.0,"min_pressure_bar":0.0,"min_temperature_C":15.0,"name":"One milliliter PFA photocoil","system":"420 nm photochemistry module","type":"photochemical coil","volume_mL":1.0,"wavelength_nm":420.0}],"safety_accessories":[{"capabilities":
["shield_or_containment","vented_separator"],"compatible_hazards":
["high_intensity_light","oxygen_enriched_organic_stream","acetonitrile_vapor"],"equipment_id":"safety_photo_enclosure","max_pressure_bar":1.0,"name":"Interlocked photochemical

enclosure and vent","type":"interlocked enclosure"}],"schema_version":"flowpilot_lab_inventory_v3.0","separators":[{"equipment_id":"separator_photo_gl","max_flow_mL_min":5.0,"max_pressure_bar":5.0,"name":"Vented gas-liquid separator","supported_phases":["gas","liquid"],"type":"gas-liquid separator"}],"strict_assignment":true,"temperature_controllers":[{"allowed_temperatures_C":[21.0,25.0],"equipment_id":"controller_photo_thermostat","max_temperature_C":40.0,"min_temperature_C":15.0,"name":"Photoreactor thermostat","type":"recirculating thermostat"}],"tubing":[{"ID_mm":1.0,"equipment_id":"tubing_photo_pfa_10","length_m":1.273,"material":"PFA","max_pressure_bar":10.0,"max_temperature_C":60.0,"name":"PFA photoreactor coil tubing","transparent":true}]}

CHEMIST-CONFIRMED CHEMISTRY IDENTITY:

{"confirmed":true,"mechanism_type":"photocatalytic aerobic oxidation","reaction_class":"oxidation","reaction_name":"photocatalytic aerobic oxidation of Fmoc-L-methionine"}

Case study 3: CuAAC

Benchmark title: Cu/C-catalyzed azide-alkyne cycloaddition

RAW BATCH PROTOCOL:

Benzyl azide (0.50 mmol, 1.0 equiv) and phenylacetylene (0.55 mmol, 1.1 equiv) were combined with Cu/C catalyst (10 mol% Cu) in acetone (2.0 mL; benzyl azide concentration 0.25 M). The homogeneous liquid reaction mixture was heated under microwave batch conditions at 150 °C for 3 min. The corresponding 1,4-disubstituted 1,2,3-triazole was isolated in 94% yield. Translate this batch procedure into one conservative continuous-flow screening design. Preserve the Cu/C catalyst strategy and explicitly report feed preparation, reactor topology, temperature, liquid flow, pressure, reactor/contact volume, residence-time basis, collection, work-up, and azide/pressure safety controls.

OBJECTIVE:

Produce one executable, inventory-constrained first flow screen without using the held-out source answer.

HARD CONSTRAINTS:

- Use only the authoritative inventory supplied below.

- Use a single premixed liquid feed through the Cu/C packed-bed cartridge.
- Do not invent a gas stream, photochemical equipment, or unlisted hardware.
- Report catalyst-contact residence time and make reactor volume, liquid flow, and residence time mathematically consistent.
- Return SCREEN unless the inventory cannot support a safe design.

AVAILABLE INVENTORY (JSON):

{"BPR_available":[20.0],"collectors":[{"equipment_id":"collector_amber_100","max_pressure_bar":1.0,"name":"Vented collection vessel","type":"collection vessel","volume_mL":100.0}],"connectors":[{"equipment_id":"connector_xcube_cart","material":"stainless steel","max_pressure_bar":40.0,"name":"X-Cube cartridge connector","supported_ID_mm":[4.0],"type":"cartridge holder"}],"degassers":[],"filters":[],"gas_hardware":[],"light_sources":[],"mixers":[],"pressure_controllers":[{"equipment_id":"bpr_xcube_20","max_pressure_bar":20.0,"min_pressure_bar":20.0,"name":"X-Cube BPR","setpoints_bar":[20.0],"type":"BPR"}],"pumps":[{"compatible_materials":["acetone"],"equipment_id":"pump_xcube_hplc","max_flow_rate_mL_min":3.0,"max_pressure_bar":40.0,"min_flow_rate_mL_min":0.1,"name":"X-Cube HPLC pump","type":"HPLC"}],"reactor_trains":[],"reactors":[{"ID_mm":4.0,"configuration":"single packed-bed cartridge","equipment_id":"reactor_cuc_cart_030","material":"stainless steel","max_pressure_bar":40.0,"max_temperature_C":200.0,"min_pressure_bar":0.0,"min_temperature_C":20.0,"name":"Cu/C CatCart packed-bed cartridge","system":"X-Cube","type":"packed-bed","volume_mL":0.3}],"safety_accessories":[{"capabilities":["shield_or_containment"],"compatible_hazards":["energetic_azide"],"equipment_id":"safety_cuaac_shield","name":"Azide-service blast shield and secondary containment","type":"shielded enclosure"}],"schema_version":"flowpilot_lab_inventory_v2.0","separators":[],"strict_assignment":true,"temperature_controllers":[{"equipment_id":"heater_xcube","max_temperature_C":200.0,"min_temperature_C":20.0,"name":"X-Cube cartridge heater","type":"heater"}],"tubing":[{"ID_mm":4.0,"equipment_id":"tubing_xcube_ss_4mm","length_m":0.06,"material":"stainless steel","max_pressure_bar":40.0,"max_temperature_C":200.0,"name":"X-Cube cartridge body","transparent":false}]}

CHEMIST-CONFIRMED CHEMISTRY IDENTITY:

{"confirmed":true,"mechanism_type":"copper-catalyzed cycloaddition","reaction_class":"cuaac","reaction_name":"copper-catalyzed azide-alkyne cycloaddition"}

Internal FlowPilot module-ablation screen

The internal screen used Qwen3.8-27B as the fixed generator and the same three chemistry cases. Fifteen architecture conditions were tested, giving 45 condition-case outcomes. Each condition-case cell was generated once; therefore, the reported standard deviations describe variation between the three chemistry cases and must not be interpreted as generation-repeat uncertainty. Disabled specialist agents received no LLM call and no default-score penalty. Active council weights were renormalized. Deterministic inventory and safety gates remained active for executable FlowPilot conditions. The full condition table is provided below for transparency; the manuscript figure may display a selected subset for readability.

Table S10. Internal FlowPilot module-ablation conditions and outcomes.

Condition	Mean score	Between-case SD	Executable	LLM calls	Critical flags
Full FlowPilot	0.928	0.027	100%	13.3	0
Without Kinetics agent	0.924	0.057	100%	12.0	0
Without Safety agent	0.915	0.060	100%	12.3	0
Without Fluidics agent	0.913	0.042	100%	10.7	0
Without Skeptic audit	0.904	0.039	100%	13.3	0
Without Chemistry agent	0.902	0.054	100%	12.0	0
One-shot	0.743	0.167	0%	1.0	14
No council	0.670	0.137	0%	3.0	19

The screen supports descriptive module attribution. It should not be interpreted as a statistically powered ranking of every internal configuration. Notably, removal of an individual specialist did not necessarily reduce the aggregate score for this small case set, whereas eliminating the complete council caused a substantial loss of executable design quality.

Table S11. Matched one-shot and FlowPilot architecture scores by generator model.

Generator model	One-shot	FlowPilot	Mean difference
Qwen3.6-27B	0.765 +/- 0.014	0.904 +/- 0.029	+0.139
Qwen3.8-27B	0.747 +/- 0.019	0.915 +/- 0.011	+0.168
GPT-4o	0.684 +/- 0.011	0.910 +/- 0.002	+0.226
Claude Sonnet 4.6	0.890 +/- 0.014	0.924 +/- 0.024	+0.034
Claude Opus 4.6	0.883 +/- 0.015	0.932 +/- 0.010	+0.049

Values are mean +/- sample SD across three repeat-level campaign means. The difference column is FlowPilot minus one-shot.

Across the 45 retained outcomes per architecture:

Table S12. Aggregate architecture-level score and critical-error summary.

Architecture	Mean candidate score	Campaigns without critical flags	Raw judge critical flags
One-shot	0.794	18/45	98
FlowPilot	0.917	43/45	4

Raw judge critical flags count separate judge reports and are not the same as the number of unique error types. The campaign-level error register consolidates judges that identified the same criterion-level error.

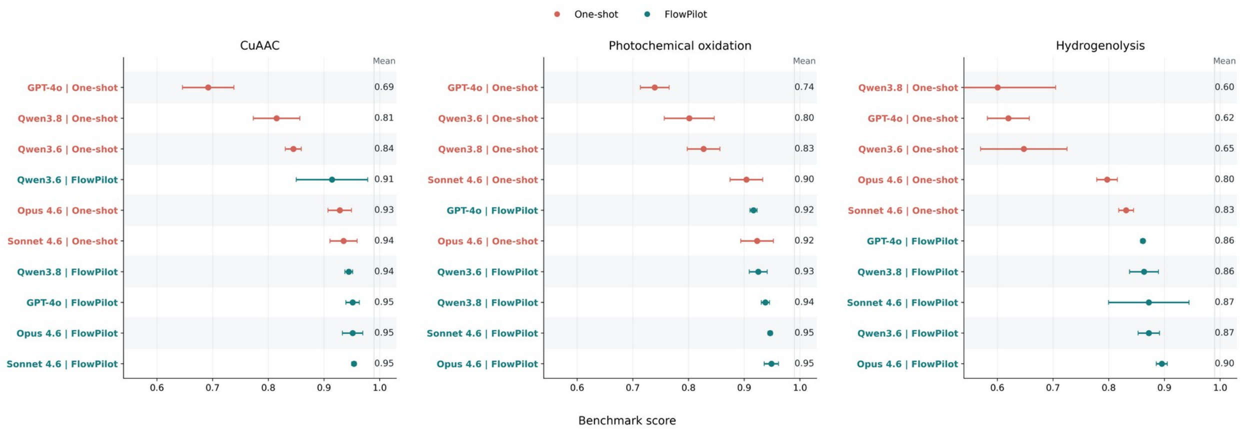

Figure S7. Ranked one-shot and FlowPilot fixed-criteria multi-LLM evaluation scores for CuAAC, photochemical oxidation, and hydrogenolysis. Points are means and error bars are sample standard deviations across three generation repeats.

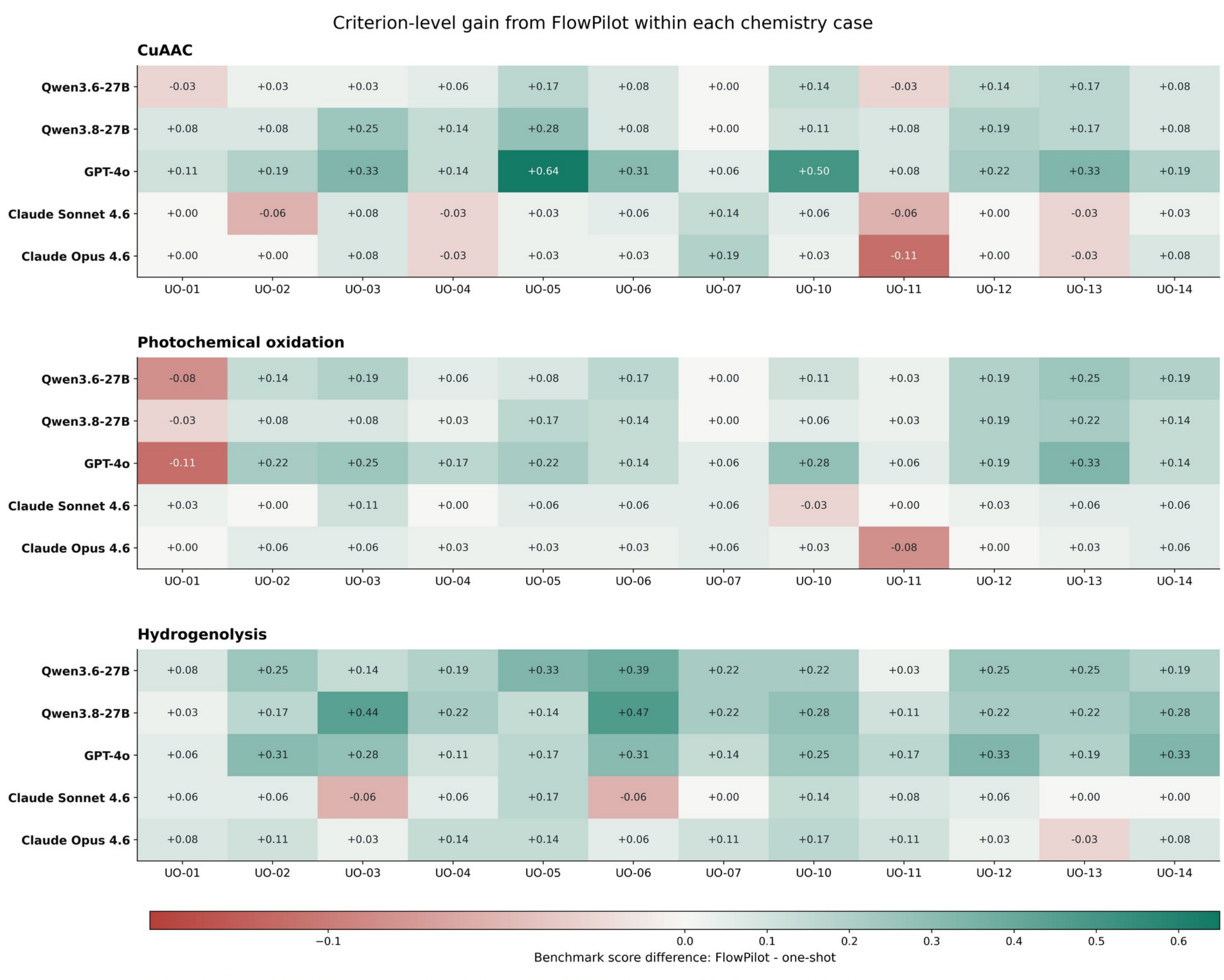

Figure S8. Mean paired FlowPilot-minus-one-shot score difference for each applicable universal criterion, separated by chemistry. Positive values favor FlowPilot. UO-08 and UO-09 are excluded where applicability is not uniform.

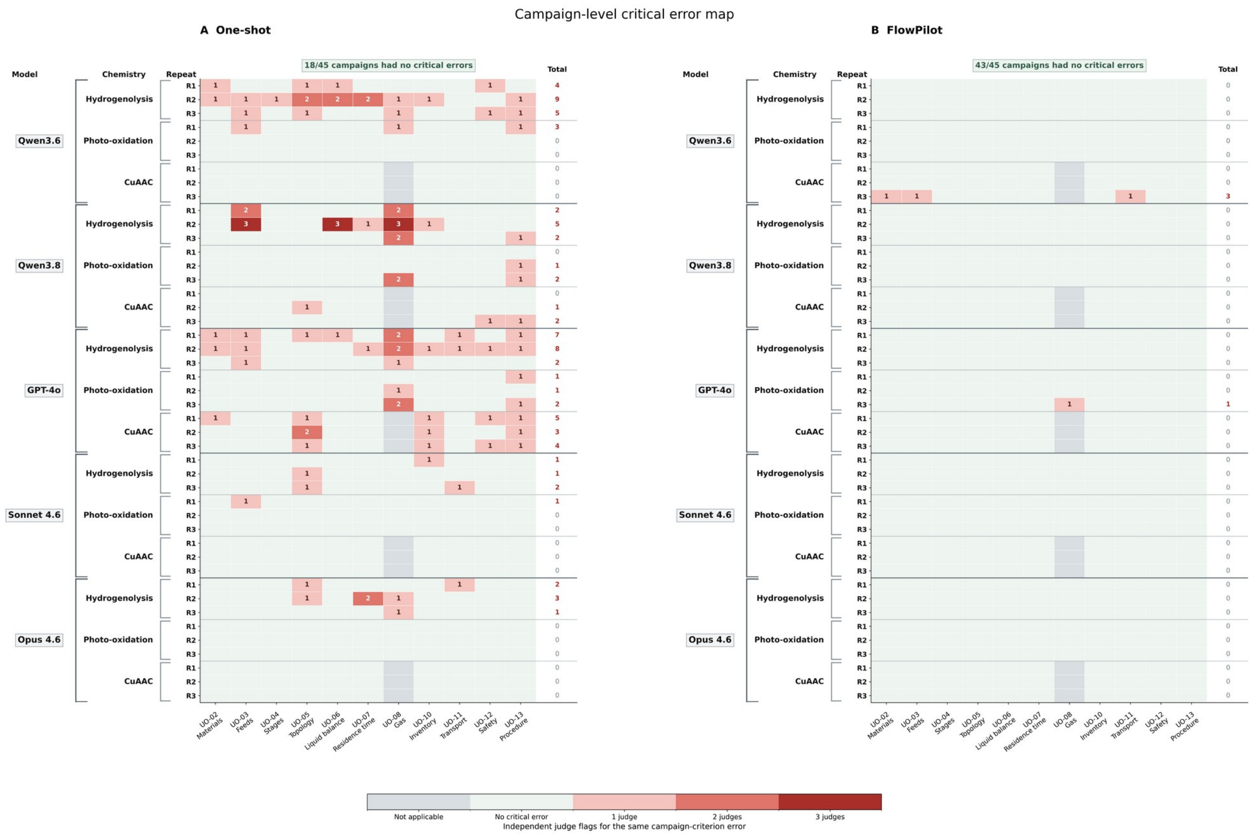

Figure S9. Critical-error criteria identified by the independent judge panel for each model, chemistry, architecture, and repeat. Cell numbers report independent judge agreement for the same campaign-criterion error; totals count distinct affected criteria.

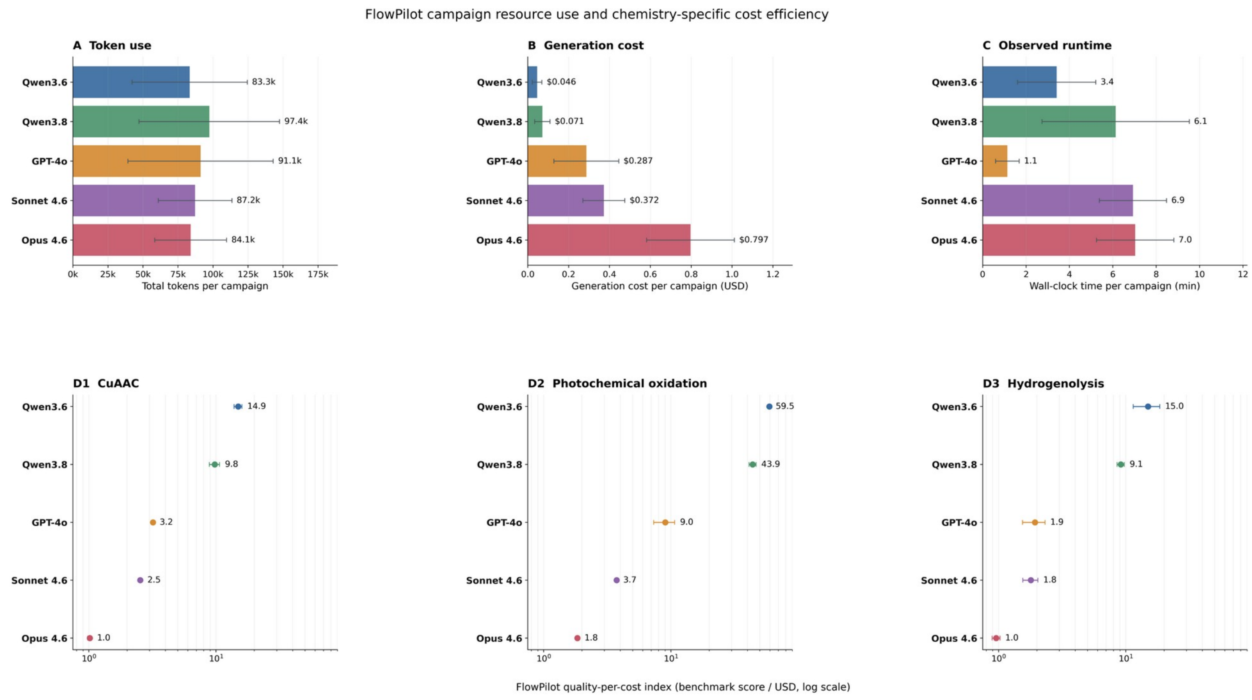

Figure S10. FlowPilot-only token use, measured generation cost, observed runtime, and chemistry-specific quality-per-cost index. Resource panels report mean and sample standard deviation across nine campaigns per model. Runtime is environment-dependent, and judge-evaluation costs are excluded.

4 Deterministic engineering, inventory realization, and refinement

FlowPilot separates model-proposed independent variables from quantities that can be recomputed. Flow rates, gas compression, molar equivalents, reactor volumes, stage times, pressure-drop indicators, heat-transfer indicators, and equipment assignments are recalculated after candidate selection and again after inventory realization. A numerical value becomes a run instruction only if the recomputed proposal, stage table, stream table, inventory allocation, process graph, and final contract agree within the implemented tolerances.

Residence-time and gas-flow conventions

Liquid molar flow is obtained from liquid concentration and volumetric flow. For a single liquid phase, nominal residence time closes as tau = V/Q. Gas feeds are stored as MFC setpoints at the declared inlet/STP reference. The corresponding in-channel volumetric flow is calculated by ideal-gas scaling, Qg,ch = Qg,STP(Tch/TSTP)(PSTP/Pabs). FlowPilot reports both the inlet/STP apparent time, V/(QL + Qg,STP), and the pressure-corrected in-channel apparent time, V/(QL + Qg,ch), when those definitions are applicable. Gas equivalents are calculated from gas molar flow and limiting-liquid molar flow; they are not inferred from a gas-to-liquid volumetric ratio. In packed beds or poorly characterized multiphase systems, contact time is not equated automatically with empty-bed space time: measured phase holdup or residence-time-distribution data are requested where required.

Table S13. Deterministic quantities, definitions, and closure checks.

Quantity	Implemented definition or basis	Required closure
Limiting molar flow	n_lim = C_lim QL	Concentration, liquid flow, and stream composition agree
Liquid residence time	tau_L = V_L/QL	Reactor liquid volume equals QL tau_L
In-channel gas flow	Qg,ch = Qg,STP(Tch/TSTP)(PSTP/Pabs)	Temperature and absolute-pressure bases are explicit
Gas equivalents	n_gas/n_lim from standard gas molar volume	Gas identity and gas fraction are included
Gas holdup estimate	epsilon_g = Qg,ch/(QL + Qg,ch)	Used only with the declared two-phase approximation
Multistage total	tau_total = sum(tau_i)	Each stage uses cumulative local inlet flow and assigned volume
Pressure drop	Single- or two-phase model selected by phase regime	Calculated drop remains within pump, tubing, reactor, and BPR limits
Thermal conductance	UA = U A_wall; Da_th = Qgen/Qrem	Assumptions, geometry, and thermal margin are reported

Multistage calculation and stream introduction

Each reactive stage receives a separate stage identifier, inlet composition, cumulative liquid and gas flow, assigned reactor volume, temperature, pressure, and residence-time basis. A feed introduced between reactors contributes only to downstream cumulative flow. Quench and work-up operations are represented in topology but are not counted as reactive stages. If only a total target time is available, allocation uses declared stage times or stage batch-time weights and otherwise applies an explicit equal-allocation fallback. Final validation checks both the stage sum and each local volume-flow-time identity.

Heat-transfer screening

The heat-transfer module reports wall area, surface-to-volume ratio, overall thermal conductance UA, estimated heat generation and removal rates, a thermal Damkohler-like ratio, and a screening margin. The implemented relations are A_wall = pi d L, Qrem = UA DeltaTlm, and Da_th = Qgen/Qrem. Reaction enthalpy, overall heat-transfer coefficient, and driving-temperature difference can be estimated from reaction and reactor classes when measurements are unavailable. These values are screening assumptions rather than calorimetric evidence; an apparently favorable estimate does not replace reaction calorimetry for scale-up or strongly exothermic chemistry.

Inventory normalization and executable topology

Laboratory inventories may be entered as structured JSON or extracted from PDF, Word, PowerPoint, spreadsheet, and text sources. Extracted content is normalized into typed equipment classes and reviewed before use. Each item carries a stable equipment_id and available capability fields. Free-text statements such as 'no inline degasser' become explicit unavailable capabilities or hard constraints rather than prompt-only notes. During design realization, every required operation is assigned to an inventory item. The allocator checks type, material, inner diameter, volume, flow range, temperature, pressure, wavelength, phase compatibility, and connectivity where declared. Under strict assignment, absent or invented hardware cannot be published in the executable topology.

Table S14. Inventory classes and deterministically checked capabilities.

Inventory class	Representative capability fields	Design role
Pumps and MFCs	Minimum/maximum flow, pressure, compatible fluids or gas identity	Quantified feed delivery
Mixers/connectors	Inputs, material, pressure, supported ID, connectivity	Stream joining and interstage connection
Reactors/trains	Type, volume, ID, material, temperature, pressure, light compatibility	Reactive residence volume
Light/temperature control	Wavelength, power, allowed temperature setpoints	Photochemical and thermal control
BPRs/tubing	Setpoints or range, pressure rating, ID, length, material	Pressure and hydraulic realization
Separators/collectors	Supported phases, flow, pressure, volume, venting	Depressurization and product collection
Safety accessories	Hazard compatibility and control capability	Purge, check valve, shielding, containment, and venting

A process diagram is generated from the same inventory-assigned topology stored in the final contract. If assignment or numerical closure fails, FlowPilot may render a requirements topology for diagnosis, but it does not label that diagram executable and does not publish numerical run instructions. This distinction prevents a polished intermediate graph from being mistaken for a laboratory-ready process.

Incremental experimental refinement

Closed-loop refinement accepts one or more ExperimentRecord objects containing actual operating conditions, measured response, and operational observations. Measured evidence has higher authority than model inference. The module diagnoses underconversion, selectivity loss, solubility or fouling, pressure instability, and gas-liquid instability; it then proposes bounded changes to residence time, liquid and gas flow, temperature, concentration, pressure, tubing, filtration, or safety controls. For multiple experiments, the campaign preserves every observation, identifies the best observed anchor, and may fit a simple apparent response-versus-residence-time model. Extrapolated targets are reported as screening estimates with acceptance criteria, not guaranteed yields. Every cycle creates a new version while retaining the preceding design and measurements.

Table S15. Experimental feedback fields and their role in refinement.

Evidence group	Recorded fields	Refinement use
Timing and geometry	Residence-time basis, inlet and channel time, flow, volume, tubing ID	Reconcile actual exposure and throughput
Gas delivery	Gas identity, MFC/STP flow, channel flow, equivalents, pressure	Recompute gas supply and apparent time bases
Chemistry response	Yield, conversion, selectivity, product and starting-material fractions	Anchor response trend and target acceptance
Operating state	Temperature, concentration, wavelength, light power	Separate kinetic, photonic, and concentration changes
Run observations	Pressure drift, clogging, precipitation, phase stability, impurity notes	Trigger feasibility and safety interventions

5 Computational provenance and interpretation boundaries

Model routing and run records

The user interface exposes separate model routes for upstream chemistry interpretation and downstream proposal/council execution. The selected provider, model identifier, and local OpenAI-compatible endpoint mapping are applied for the duration of one serialized run and restored afterward. Model routing is written to pipeline-runtime provenance. Because hosted model aliases and local checkpoints can change, a reproducible report should retain the exact model identifier, endpoint or provider, generation parameters, software revision, dependency environment, frozen

DesignInputPackage, inventory profile, raw model events, deterministic audit, and canonical final result.

Table S16. Minimum records required for audit and computational replay.

Record	Purpose	Typical stored object
Frozen input	Reconstruct chemist-provided facts and authority	DesignInputPackage and inventory profile
Model configuration	Identify stochastic generators and routes	Provider, model ID, endpoint, temperature, seed where supported
Prompts/events	Audit module inputs and raw responses	Timestamped LLM event log and token telemetry
Deterministic state	Verify calculations and gates	Calculation sheet, validation report, allocation, topology
Canonical output	Identify the published design	FinalDesignContract, content hash, rendered artifacts
Software environment	Support replay of code-owned behavior	Git revision, Python/dependency versions, configuration

Statistical and evidential limitations

The architecture comparison uses three generation repeats for each retained model-case-architecture cell. Error bars therefore describe observed generation-repeat variation for this frozen benchmark, not population confidence intervals. LLM judges apply a fixed rubric and are identity-masked, but they are not substitutes for blinded flow-chemist assessment. Deterministic verification reduces arithmetic ambiguity but does not validate unknown kinetics, selectivity, catalyst lifetime, phase holdup, or scale-dependent transport. Benchmark scores measure the completeness and defensibility of proposed screening designs. Experimental execution and measured chemical performance remain the final validation.

REFERENCES

No references are cited exclusively in the Supporting Information; literature citations are provided in the main article.